

Qatar-2b

R-0963

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_170327 · created 2026-07-29T16:28:44+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457839.86983 +/- 0.00042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.019 +/- 0.025
Transit depth (Rp/Rs)^2	0.037 +/- 0.095 [%]
Orbital Inclination (inc)	89.79 +/- 1.74
Airmass coefficient 1 (a1)	1.036 +/- 0.021
Airmass coefficient 2 (a2)	-0.026 +/- 0.029
Scatter in the residuals of the lightcurve fit is	1.99 %
Best Comparison Star	None
Optimal Aperture	9.79
Optimal Annulus	13.05
Transit Duration (day)	0.0666 +/- 0.0024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.019 ± 0.025 vs 0.16327 (deviation 5.11σ)
Duration fit vs published	0.0666 d vs 0.0767 d (deviation 3.72σ)
Mid-time O–C	0.9 min
Depth SNR	0.7
Residual scatter	1.99 %

Light curve

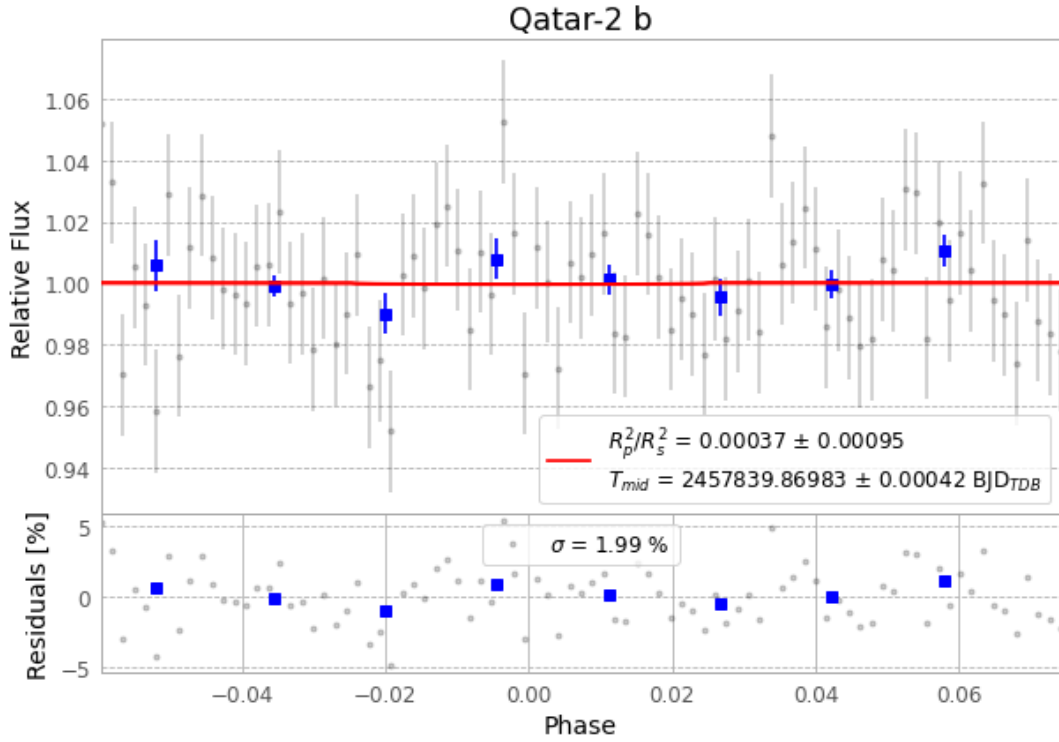


Figure 1 — FinalLightCurve_Qatar-2 b_2017-03-26.png (SHA-256 d92e3f03f8a4ed59...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [274, 451], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ee1db4eb522b613d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

87 FITS frames (57 MB), Qatar-2170327064815.FITS → Qatar-2170327110612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-170327.log (5033fe368eeb...), FinalLightCurve_Qatar-2 b_2017-03-26.png (d92e3f03f8a4...), FinalLightCurve_Qatar-2 b_2017-03-26.csv (1f6e7ce85d1c...)

Compute resources

Wall time 310.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 16:28Z.